

Multi-Photon Events in a Cavity-Based Atom-Photon Gate

Stephan Dürr

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We consider a CPHASE gate similar to Refs. [1]. A single atom is located in an optical resonator. The Hilbert space of the two-level atom has the computational basis ($|\uparrow\rangle, |\downarrow\rangle$), representing the upper and lower hyperfine state of the electronic ground state. Ideally, a single-photon Fock state would be perfectly reflected off the cavity. Its polarization state has a Hilbert space with the orthonormal basis ($|H\rangle, |V\rangle$) of horizontal and vertical polarization. We denote the corresponding single-photon Fock states as $|1_H\rangle = |1\rangle_H$ and $|1_V\rangle = |1\rangle_V$. Ideally, this reflection should be a CPHASE gate, i.e. a bilinear map characterized by

$$|\uparrow, 1_H\rangle \mapsto e^{i\pi} |\uparrow, 1_H\rangle, \quad |\downarrow, 1_H\rangle \mapsto |\downarrow, 1_H\rangle, \quad |\uparrow, 1_V\rangle \mapsto |\uparrow, 1_V\rangle, \quad |\downarrow, 1_V\rangle \mapsto |\downarrow, 1_V\rangle. \quad (1)$$

In the following, we study how multi-photon events and detector dark counts deteriorate the postselected Bell-state fidelity of the CPHASE gate.

1 Multi-Photon Events

In an experiment, one often uses an attenuated light pulse generated from a laser and combines it with postselection upon detection of exactly one photon to avoid the need for a single-photon source. This system can be described by the methods of Ref. [2] provided that the reflection off the cavity is well-approximated by a *linear-response* model. This is the case in the weak driving limit, because here the nonlinearity of the Jaynes-Cummings ladder is not explored. This regime is characterized by the condition that the probability that two photons are *inside the cavity simultaneously* is negligible. According to the discussion above Eq. (9) in Ref. [2], the latter condition is equivalent to $|\alpha_{\text{in}}|^2 \ll 2\kappa T_{\text{in}}$, where T_{in} is the duration of the incoming light pulse, 2κ is the $1/e$ lifetime of a resonant photon, if it were prepared inside the cavity in the absence of atoms, and α_{in} is the amplitude of the incoming coherent state.

1.1 Cavity-Output State

Assuming that the conditional π phase shift applies to each photon, we ideally obtain, see Eq. (3) of Ref. [2] for H polarization

$$|\uparrow, \alpha_H\rangle \mapsto |\uparrow, -\alpha_H\rangle, \quad |\downarrow, \alpha_H\rangle \mapsto |\downarrow, \alpha_H\rangle, \quad |\uparrow/\downarrow, \alpha_V\rangle \mapsto |\uparrow/\downarrow, \alpha_V\rangle, \quad (2)$$

where $|\alpha_H\rangle = |\alpha\rangle_H$ etc. Note that the reflected amplitude for the polarization V is independent of the atomic state because light with this polarization does not interact with the atom. The value of the reflected amplitude for V polarization is the same as for the initial state $|\downarrow, \alpha_H\rangle$ because for this internal state, the light also does not interact with the atom.

Now, we take into account that some photons might be lost in the cavity. As we consider linear response, the loss results in a reflected coherent state with amplitude $r_{\uparrow/\downarrow}$ if the light is H polarized and the atomic state is $|\uparrow\rangle$ and $|\downarrow\rangle$, respectively. If the light is V polarized, then the reflected amplitude is always $r_{\downarrow}$, i.e., we generalize Eq. (2) to¹

$$|\uparrow, \alpha_{\text{in},H}\rangle \mapsto |\uparrow, r_{\uparrow,H}\rangle, \quad |\downarrow, \alpha_{\text{in},H}\rangle \mapsto |\downarrow, r_{\downarrow,H}\rangle, \quad |\uparrow/\downarrow, \alpha_{\text{in},V}\rangle \mapsto |\uparrow/\downarrow, r_{\downarrow,V}\rangle, \quad (3)$$

where $|r_{\uparrow,H}\rangle = |r_{\uparrow}\rangle_H$ etc. Typically, $\arg(-r_{\uparrow}) \approx 0$ and $\arg(r_{\downarrow}) \approx 0$. We recover Eq. (2) for $r_{\uparrow}/\alpha_{\text{in}} = -1$ and $r_{\downarrow}/\alpha_{\text{in}} = 1$.

The lost light emerges in modes orthogonal to the reflected mode. As we consider linear response, the lost light is in a coherent state with amplitude $\ell_{\uparrow/\downarrow}$. Here $|\ell_{\uparrow}\rangle = |a_{\uparrow}, m_{\uparrow}\rangle$ is a product of light lost from the atom and from the mirrors, see Eq. (15) in Ref. [2]. If the atom is uncoupled, then the light is only lost from the mirrors $|\ell_{\downarrow}\rangle = |a_{\downarrow}, m_{\downarrow}\rangle$ with

$$a_{\downarrow} = 0. \quad (4)$$

Note that the polarization of the light lost from the mirrors depends on the polarization of the incoming light. We assume that scattering off the mirror is a unitary process. Hence, with the two incoming polarization modes H and V being orthogonal, the light scattered off the mirrors $|m_{\uparrow/\downarrow,H}\rangle = |m_{\uparrow/\downarrow}\rangle_H$ and $|m_{\downarrow,V}\rangle = |m_{\downarrow}\rangle_V$ are also in orthogonal modes. Including the lost fields, Eq. (3) becomes

$$|\uparrow, \alpha_{\text{in},H}\rangle \mapsto |\uparrow, r_{\uparrow,H}, a_{\uparrow}, m_{\uparrow,H}\rangle, \quad (5a)$$

$$|\downarrow, \alpha_{\text{in},H}\rangle \mapsto |\downarrow, r_{\downarrow,H}, m_{\downarrow,H}\rangle, \quad (5b)$$

$$|\uparrow/\downarrow, \alpha_{\text{in},V}\rangle \mapsto |\uparrow/\downarrow, r_{\downarrow,V}, m_{\downarrow,V}\rangle, \quad (5c)$$

where, as often in the literature, we omitted modes, which are in their vacuum state.

We consider the input state

$$|\psi_{\text{in}}\rangle = |\alpha_{\text{in},D}\rangle \otimes \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}} \quad (6)$$

with the polarization state

$$|D\rangle = \frac{|H\rangle + |V\rangle}{\sqrt{2}}. \quad (7)$$

Hence²

$$|\alpha_{\text{in},D}\rangle = \left| \frac{\alpha_{\text{in}}}{\sqrt{2}}H, \frac{\alpha_{\text{in}}}{\sqrt{2}}V \right\rangle. \quad (8)$$

¹If we wrote $r_c = r_{\uparrow}$ and $r_u = r_{\downarrow}$ for the coupled and uncoupled case, the notation might be a bit more intuitive. However, aiming at compatibility with the notation of Ref. [2], we prefer $r_{\uparrow/\downarrow}$.

²To see this, note that the creation operators are related by $a_D^{\dagger} = \frac{1}{\sqrt{2}}(a_H^{\dagger} + a_V^{\dagger})$ with $[a_H, a_V^{\dagger}] = [a_H, a_V] = 0$. Hence, the displacement operator $D_D(\alpha) = \exp(\alpha a_D^{\dagger} - \alpha^* a_D)$ obeys $D_D(\alpha) = D_H(\frac{\alpha}{\sqrt{2}})D_V(\frac{\alpha}{\sqrt{2}})$. Hence $|\alpha_D\rangle = D_D(\alpha)|0\rangle = |\frac{\alpha_{\text{in}}}{\sqrt{2}}H, \frac{\alpha_{\text{in}}}{\sqrt{2}}V\rangle$. This is much like in a beam splitter.

The state after the interaction of the light and the cavity reads

$$|\psi_{\text{ent}}\rangle = \frac{|\uparrow, \frac{r_{\uparrow}}{\sqrt{2}}H, \frac{r_{\downarrow}}{\sqrt{2}}V, \frac{a_{\uparrow}}{\sqrt{2}}, \frac{m_{\uparrow}}{\sqrt{2}}H, \frac{m_{\downarrow}}{\sqrt{2}}V\rangle + |\downarrow, \frac{r_{\downarrow}}{\sqrt{2}}H, \frac{r_{\uparrow}}{\sqrt{2}}V, \frac{m_{\downarrow}}{\sqrt{2}}H, \frac{m_{\uparrow}}{\sqrt{2}}V\rangle}{\sqrt{2}}. \quad (9)$$

Here, the system is entangled with the environment of lost photons. To obtain Eq. (9), we again used the assumption of weak drive. This implies that the probability of a photon with polarization H and a photon with polarization V being in the cavity *simultaneously* is negligible. Hence, each polarization is reflected independently of the presence of the other polarization.

Note that in analogy Eq. (8)

$$|r_{\downarrow,D}\rangle = \left| \frac{r_{\downarrow}}{\sqrt{2}}H, \frac{r_{\downarrow}}{\sqrt{2}}V \right\rangle. \quad (10)$$

We take the partial trace over the environment consisting of the lost photons. We obtain the cavity-output state

$$\rho_{\text{out}} = \frac{|\uparrow, \frac{r_{\uparrow}}{\sqrt{2}}H, \frac{r_{\downarrow}}{\sqrt{2}}V\rangle\langle\uparrow, \frac{r_{\uparrow}}{\sqrt{2}}H, \frac{r_{\downarrow}}{\sqrt{2}}V| + |\downarrow, r_{\downarrow,D}\rangle\langle\downarrow, r_{\downarrow,D}|}{2} + V_{\text{out}} \frac{e^{i\theta} |\downarrow, r_{\downarrow,D}\rangle\langle\uparrow, \frac{r_{\uparrow}}{\sqrt{2}}H, \frac{r_{\downarrow}}{\sqrt{2}}V| + \text{H.c.}}{2}. \quad (11)$$

Using the methods of Ref. [2], we find that the visibility $V_{\text{out}} \in [0, 1]$ and the phase $\theta \in]-\pi, \pi]$ are given by

$$V_{\text{out}} e^{i\theta} = \left\langle \frac{a_{\uparrow}}{\sqrt{2}}, \frac{m_{\uparrow}}{\sqrt{2}}H, \frac{m_{\downarrow}}{\sqrt{2}}V \left| \frac{m_{\downarrow}}{\sqrt{2}}H, \frac{m_{\downarrow}}{\sqrt{2}}V \right\rangle = \left\langle \frac{a_{\uparrow}}{\sqrt{2}} \left| 0 \right\rangle \left\langle \frac{m_{\uparrow}}{\sqrt{2}}H \left| \frac{m_{\downarrow}}{\sqrt{2}}H \right\rangle \underbrace{\left\langle \frac{m_{\downarrow}}{\sqrt{2}}V \left| \frac{m_{\downarrow}}{\sqrt{2}}V \right\rangle}_{=1} \right. \right. \quad (12)$$

Equation (12) shows that only the photons lost for the input polarization H contribute to the visibility V_{out} . Hence, we would obtain the same V_{out} if we considered an incoming field with only H polarization with amplitude $\alpha_{\text{in}}/\sqrt{2}$. That situation was already considered in Ref. [2]. Hence, we can use Eq. (30) of Ref. [2] to calculate the the visibility

$$V_{\text{out}} = \exp \left(-2 \frac{L_{\text{gen}}}{1 - L_{\text{gen}}} \alpha_{\text{out}}^2 \right), \quad (13)$$

where $\alpha_{\text{out}} = \frac{1}{2} \left| \frac{r_{\uparrow}}{\sqrt{2}} - \frac{r_{\downarrow}}{\sqrt{2}} \right|$ and where, see the discussion below Eq. (36) of Ref. [2]

$$L_{\text{gen}} = 1 - \eta_{\text{esc}} \frac{C}{1 + C}, \quad (14)$$

where $\eta_{\text{esc}} = \kappa_{\text{in}}/\kappa$ is the probability that a photon prepared in the cavity in the absence of atoms leaves the cavity through the input-output coupler, and where³

$$C = \frac{|g|^2}{\kappa\gamma} \quad (15)$$

³Caveat: In some parts of the literature, C is defined to be $\frac{|g|^2}{2\kappa\gamma}$.

is the cooperativity.

In addition, according to Eq. (9a) of Ref. [2] we obtain

$$\frac{r_{\uparrow/\downarrow}}{\alpha_{\text{in}}} = -1 + \frac{2\eta_{\text{esc}}}{1 + C/\Lambda_{\uparrow/\downarrow}^2} = \begin{cases} -1 + \frac{2\eta_{\text{esc}}}{1 + C}, & \text{for } r_{\uparrow} \\ -1 + 2\eta_{\text{esc}}, & \text{for } r_{\downarrow}, \end{cases} \quad (16)$$

where we used $\Lambda_{\uparrow} = 1$, see Eq. (12) of Ref. [2], and $\Lambda_{\downarrow} = \infty$, see the discussion below Eq. (36) of Ref. [2]. Moreover, using Eqs. (5), (9b), (9c), and (25) of Ref. [2], we find

$$\theta = \arg(\langle \ell_{\uparrow} | \ell_{\downarrow} \rangle) = \arg(e^{a_{\uparrow}^* a_{\downarrow} + m_{\uparrow}^* m_{\downarrow}}) = \arg(e^{i \text{Im}(a_{\uparrow}^* a_{\downarrow} + m_{\uparrow}^* m_{\downarrow})}) = 0. \quad (17)$$

1.2 State in Front of the Detector

We assume that only a fraction $1 - L_{\text{aft}}$ of the photons reflected off the cavity are detected. In doing so, we include loss in optical components after the cavity as well as the non-unity detector efficiency. The detector efficiency can be modeled as additional optical loss followed by a hypothetical, lossless detector. Using the methods of Ref. [2] one easily finds that the state in front of this hypothetical, lossless detector is given by

$$\rho_{\text{det}} = \frac{|\uparrow, \frac{d_{\uparrow}}{\sqrt{2}}H, \frac{d_{\downarrow}}{\sqrt{2}}V\rangle\langle\uparrow, \frac{d_{\uparrow}}{\sqrt{2}}H, \frac{d_{\downarrow}}{\sqrt{2}}V| + |\downarrow, d_{\downarrow,D}\rangle\langle\downarrow, d_{\downarrow,D}|}{2} + V_{\text{det}} \frac{e^{i\theta} |\downarrow, d_{\downarrow,D}\rangle\langle\uparrow, \frac{d_{\uparrow}}{\sqrt{2}}H, \frac{d_{\downarrow}}{\sqrt{2}}V| + \text{H.c.}}{2} \quad (18)$$

with

$$\begin{aligned} V_{\text{det}} &= V_{\text{out}} \left\langle \frac{\rho_{\text{bs}} r_{\uparrow}}{\sqrt{2}} H, \frac{\rho_{\text{bs}} r_{\downarrow}}{\sqrt{2}} V \left| \frac{\rho_{\text{bs}} r_{\downarrow}}{\sqrt{2}} H, \frac{\rho_{\text{bs}} r_{\downarrow}}{\sqrt{2}} V \right. \right\rangle \\ &= V_{\text{out}} \left\langle \frac{\rho_{\text{bs}} r_{\uparrow}}{\sqrt{2}} H \left| \frac{\rho_{\text{bs}} r_{\downarrow}}{\sqrt{2}} H \right. \right\rangle \underbrace{\left\langle \frac{\rho_{\text{bs}} r_{\downarrow}}{\sqrt{2}} V \left| \frac{\rho_{\text{bs}} r_{\downarrow}}{\sqrt{2}} V \right. \right\rangle}_{=1}, \end{aligned} \quad (19)$$

where $|\rho_{\text{bs}}|^2 = L_{\text{aft}}$. In analogy to the above, V_{det} depends only on the properties of the H polarized light and we can use the methods of Ref. [2] to obtain

$$V_{\text{det}} = \exp \left(-2 \frac{L_{\text{tot}}}{1 - L_{\text{tot}}} \alpha_{\text{det}}^2 \right) \quad (20)$$

with a total loss coefficient

$$L_{\text{tot}} = 1 - (1 - L_{\text{gen}})(1 - L_{\text{aft}}) \quad (21)$$

and with the amplitudes in front of the hypothetical, lossless detector

$$d_{\uparrow/\downarrow} = \sqrt{1 - L_{\text{aft}}} r_{\uparrow/\downarrow}, \quad \alpha_{\text{det}} = \frac{1}{2} \left| \frac{d_{\uparrow}}{\sqrt{2}} - \frac{d_{\downarrow}}{\sqrt{2}} \right|. \quad (22)$$

1.3 Bell-State Fidelity without Postselection

If the input state was a single-photon Fock state $|1\rangle_D \otimes \frac{|\uparrow\rangle + |\downarrow\rangle}{\sqrt{2}}$ and the complete system was lossless, i.e., Eq. (1) held, then the ideal output state would be a Bell state

$$|\psi_{\text{id}}\rangle = \frac{|\downarrow, 1_H\rangle + |\downarrow, 1_V\rangle - |\uparrow, 1_H\rangle + |\uparrow, 1_V\rangle}{2} = \frac{|\downarrow, 1_D\rangle - |\uparrow, 1_A\rangle}{\sqrt{2}}, \quad (23)$$

where we denote the polarization state

$$|A\rangle = \frac{|H\rangle - |V\rangle}{\sqrt{2}}. \quad (24)$$

To obtain Eq. (23), we used that⁴

$$|1_D\rangle = \frac{|1_H\rangle + |1_V\rangle}{\sqrt{2}}, \quad |1_A\rangle = \frac{|1_H\rangle - |1_V\rangle}{\sqrt{2}} \quad (25)$$

The state ρ_{det} has a Bell-state fidelity of

$$F = \langle \psi_{\text{id}} | \rho_{\text{det}} | \psi_{\text{id}} \rangle. \quad (26)$$

Now, we use

$$\begin{aligned} \left\langle 1_A \left| \frac{d_{\uparrow}}{\sqrt{2}}_H, \frac{d_{\downarrow}}{\sqrt{2}}_V \right. \right\rangle &= \frac{\langle 1_H | \frac{d_{\uparrow}}{\sqrt{2}}_H \rangle \langle 0_V | \frac{d_{\downarrow}}{\sqrt{2}}_V \rangle - \langle 0_H | \frac{d_{\uparrow}}{\sqrt{2}}_H \rangle \langle 1_V | \frac{d_{\downarrow}}{\sqrt{2}}_V \rangle}{\sqrt{2}} \\ &= \frac{d_{\uparrow} - d_{\downarrow}}{2} e^{-(|d_{\uparrow}|^2 + |d_{\downarrow}|^2)/4}, \end{aligned} \quad (27)$$

where we used

$$\langle n | \alpha \rangle = \frac{1}{\sqrt{n!}} \alpha^n e^{-|\alpha|^2/2}. \quad (28)$$

This also gives $\langle 1_D | d_{\downarrow, D} \rangle = d_{\downarrow} e^{-|d_{\downarrow}|^2/2}$. Using $\langle \uparrow | \downarrow \rangle = 0$, we find

$$\begin{aligned} F &= \frac{|\frac{1}{2}(d_{\uparrow} - d_{\downarrow})e^{-(|d_{\uparrow}|^2 + |d_{\downarrow}|^2)/4}|^2 + |d_{\downarrow}e^{-|d_{\downarrow}|^2/2}|^2}{4} \\ &\quad - V_{\text{det}} \frac{e^{i\theta} d_{\downarrow} e^{-|d_{\downarrow}|^2/2} \frac{1}{2}(d_{\uparrow}^* - d_{\downarrow}^*) e^{-(|d_{\uparrow}|^2 + |d_{\downarrow}|^2)/4} + \text{c.c.}}{4}, \end{aligned} \quad (29)$$

which we rewrite as

$$\boxed{F = \frac{|d_{\uparrow} - d_{\downarrow}|^2 e^{-|d_{\uparrow}|^2/2} + 4|d_{\downarrow}|^2 e^{-|d_{\downarrow}|^2/2}}{16} e^{-|d_{\downarrow}|^2/2} - V_{\text{det}} \frac{e^{i\theta} d_{\downarrow} (d_{\uparrow}^* - d_{\downarrow}^*) + \text{c.c.}}{8} e^{-(|d_{\uparrow}|^2 + 3|d_{\downarrow}|^2)/4}} \quad (30)$$

The minus sign in front of V_{det} comes from the minus sign in front of $|\uparrow, 1_A\rangle$ in Eq. (23),

For $d_{\downarrow} = -d_{\uparrow}$, this simplifies to $F = \frac{1}{2}(1 + V_{\text{det}} \cos \theta) |d_{\downarrow}|^2 e^{-|d_{\downarrow}|^2}$. The factor $|d_{\downarrow}|^2 e^{-|d_{\downarrow}|^2}$ results from a Poisson distribution. The factor $\frac{1}{2}(1 + V_{\text{det}} \cos \theta)$ is a sinusoidal fringe pattern with visibility V_{det} .

⁴To see this, note that the creation operators are related by $a_A^\dagger = \frac{1}{\sqrt{2}}(a_H^\dagger - a_V^\dagger)$. Hence $|1_A\rangle = a_A^\dagger |0\rangle = \frac{a_H^\dagger - a_V^\dagger}{\sqrt{2}} |0\rangle = \frac{|1_H\rangle - |1_V\rangle}{\sqrt{2}}$.

1.4 Postselected Fidelity

We consider the subensemble of events obtained by postselecting upon detecting exactly one photon. We assume that our hypothetical, lossless detector is photon-number resolving.⁵ The orthogonal projector onto the subspace with exactly one photon is

$$P_1 = |1_D\rangle\langle 1_D| + |1_A\rangle\langle 1_A| = |1_H\rangle\langle 1_H| + |1_V\rangle\langle 1_V| \quad (31)$$

with $P_1^\dagger = P_1^2 = P_1$ and $\text{tr}(P_1) = 2$. The postselected density matrix is

$$\rho_{\text{ps}} = \frac{P_1 \rho_{\text{det}} P_1}{p_1}, \quad p_1 = \text{tr}(\rho_{\text{det}} P_1). \quad (32)$$

Here, p_1 is the probability of detecting exactly one photon. We will not need an explicit expression for ρ_{ps} below. But we do need p_1 . Using $\langle \uparrow | \downarrow \rangle = 0$ and $\langle 1_A | d_{\downarrow, D} \rangle = \langle 1_A | 0_A \rangle \langle 0_D | d_{\downarrow, D} \rangle = 0$, we obtain

$$p_1 = \frac{|\langle 1_H | \frac{d_\uparrow}{\sqrt{2}} H, \frac{d_\downarrow}{\sqrt{2}} V \rangle|^2 + |\langle 1_V | \frac{d_\uparrow}{\sqrt{2}} H, \frac{d_\downarrow}{\sqrt{2}} V \rangle|^2 + |\langle 1_D | d_{\downarrow, D} \rangle|^2}{2}. \quad (33)$$

Now, using $\langle 1_H | \frac{d_\uparrow}{\sqrt{2}} H, \frac{d_\downarrow}{\sqrt{2}} V \rangle = \frac{d_\uparrow}{\sqrt{2}} e^{-(|d_\uparrow|^2 + |d_\downarrow|^2)/4}$ and $\langle 1_V | \frac{d_\uparrow}{\sqrt{2}} H, \frac{d_\downarrow}{\sqrt{2}} V \rangle = \frac{d_\downarrow}{\sqrt{2}} e^{-(|d_\uparrow|^2 + |d_\downarrow|^2)/4}$, see Eq. (27), we obtain

$$p_1 = \frac{(|d_\uparrow|^2 + |d_\downarrow|^2) e^{-|d_\uparrow|^2/2} + 2|d_\downarrow|^2 e^{-|d_\downarrow|^2/2}}{4} e^{-|d_\downarrow|^2/2}. \quad (34)$$

For the special case $|d_\downarrow| = |d_\uparrow|$, this simplifies to $p_1 = |d_\downarrow|^2 e^{-|d_\downarrow|^2}$, which is the result of a Poisson distribution.

The Bell-state fidelity of the postselected state is

$$F_{\text{ps}} = \langle \psi_{\text{id}} | \rho_{\text{ps}} | \psi_{\text{id}} \rangle \quad (35)$$

Using $P_1 |\psi_{\text{id}}\rangle = |\psi_{\text{id}}\rangle$ and Eq. (32), we find $F_{\text{ps}} = \frac{1}{p_1} \langle \psi_{\text{id}} | \rho_{\text{det}} | \psi_{\text{det}} \rangle$. Hence

$$F_{\text{ps}} = \frac{F}{p_1}. \quad (36)$$

For $d_\downarrow = -d_\uparrow$, this simplifies to

$$F_{\text{ps}} = \frac{1}{2} (1 + V_{\text{det}} \cos \theta). \quad (37)$$

For $V_{\text{det}} = 1$ and $\theta = 0$, this reaches $F = 1$.

⁵As long as the mean temporal separation between photons arriving at the detector is long compared to the detector dead time, this a good approximation.

2 Detector Dark Counts

2.1 Fidelity for Dark-Count Events

Now, we include the probability P_{dark} of the photon detection setup (as a whole) producing a single dark count in the measurement time window in a single repetition of the experiment. More precisely, the setup contains two detectors, one for each polarization. For simplicity, we assume that the dark count rates of these two detectors are identical.⁶

If a dark count occurs then this event can contribute to the postselected subensemble, only if 0 actual photon counts occur in the same shot. The corresponding orthogonal projection operator reads

$$P_0 = |0\rangle\langle 0|, \quad (38)$$

where $|0\rangle$ is the vacuum state. Obviously $P_0^2 = P_0^\dagger = P_0$ and $\text{tr}(P_0) = 1$. The corresponding postselected density matrix is

$$\rho_0 = \frac{P_0 \rho_{\text{det}} P_0}{p_0}, \quad p_0 = \text{tr}(\rho_{\text{det}} P_0). \quad (39)$$

Obviously, this results in

$$\rho_0 = \rho_0^{\text{atom}} \otimes P_0 \quad (40)$$

with some density matrix ρ_0^{atom} acting on the atomic subsystem only. Here we used $\text{tr}(P_0) = 1$. We will not need an explicit expression for ρ_0^{atom} below. But we do need p_0 . Using $\langle \uparrow | \downarrow \rangle = 0$, we obtain

$$p_0 = \frac{|\langle 0 | \frac{d_\downarrow}{\sqrt{2}} \rangle \langle 0 | \frac{d_\uparrow}{\sqrt{2}} \rangle|^2 + |\langle 0 | d_\downarrow \rangle|^2}{2}. \quad (41)$$

Hence

$$p_0 = \frac{e^{-|d_\uparrow|^2/2} + e^{-|d_\downarrow|^2/2}}{2} e^{-|d_\downarrow|^2/2}. \quad (42)$$

For $|d_\downarrow| = |d_\uparrow|$, this simplifies to $p_0 = e^{-|d_\downarrow|^2}$, which is the result of a Poisson distribution.

As we assumed that both photodetectors have the same dark count rate, the photon clicks will look like they would stem from the photonic density matrix

$$\frac{1}{2} P_1, \quad (43)$$

where we used $\text{tr}(P_1) = 2$. As the outcome of the measurement on the atom is uncorrelated from the question, which detector produced the dark count, the corresponding coincidence detection events will look like stemming from the state

$$\rho_{\text{dark}} = \rho_0^{\text{atom}} \otimes \frac{P_1}{2}. \quad (44)$$

⁶If this should not be the case, one could enforce it to become the case effectively in the lab. To explain how this can be achieved, let us denote a detector setting in which one (the other) detector measures horizontally (vertically) polarized photons as HV . Now, instead of only measuring with the basis setting HV for some time, spend one (the other) half of the time measuring in the basis setting HV (VH).

η_{esc}	$\frac{g^2}{\kappa\gamma}$	$1 - L_{\text{aft}}$	μ
0.85	$3.28 = 2 \times 1.64$	0.742	10^{-4} or 10^{-5}

Table 1: Parameters used for the numerical example here. η_{esc} is the probability that a photon prepared in the cavity in the absence of atoms leaves the cavity through the input-output coupler, $\frac{g^2}{\kappa\gamma}$ or $\frac{g^2}{2\kappa\gamma}$ are referred to as the cooperativity in the literature, only a fraction $1 - L_{\text{aft}}$ of the photons reflected off the cavity are detected, and μ is the mean number of detector dark counts (summed over both detectors) in the detection time window.

To calculate the Bell-state fidelity F_{dark} in this subensemble, we use $\rho_{\text{dark}}^\dagger = \rho_{\text{dark}}$ and $|\psi_{\text{id}}\rangle$ from Eq. (23). We find

$$F_{\text{dark}} = \langle \psi_{\text{id}} | \rho_{\text{dark}} | \psi_{\text{id}} \rangle = \frac{1}{2} \text{tr} [\rho_{\text{dark}} (|\uparrow, 1_D\rangle\langle\uparrow, 1_D| + |\downarrow, 1_A\rangle\langle\downarrow, 1_A|)] - \frac{1}{2} [\text{tr} (\rho_{\text{dark}} |\uparrow, 1_D\rangle\langle\downarrow, 1_A|) + \text{c.c.}]. \quad (45)$$

Next, we use $(A_1 \otimes B_1)(A_2 \otimes B_2) = (A_1 A_2) \otimes (B_1 B_2)$, $\text{tr}(A \otimes B) = \text{tr}(A) \text{tr}(B)$, $P_1 |1_D\rangle = |1_D\rangle$, and $P_1 |1_A\rangle = |1_A\rangle$. In addition, we use $|\uparrow, 1_D\rangle\langle\uparrow, 1_D| = |\uparrow\rangle\langle\uparrow| \otimes |1_D\rangle\langle 1_D|$ etc. We find

$$F_{\text{dark}} = \frac{1}{4} \text{tr}(\rho_0^{\text{atom}} |\uparrow\rangle\langle\uparrow|) \underbrace{\text{tr}(P_0 |1_D\rangle\langle 1_D|)}_{=1} + \frac{1}{4} \text{tr}(\rho_0^{\text{atom}} |\downarrow\rangle\langle\downarrow|) \underbrace{\text{tr}(P_0 |1_A\rangle\langle 1_A|)}_{=1} - \frac{1}{4} [\text{tr}(\rho_0^{\text{atom}} |\uparrow\rangle\langle\downarrow|) \underbrace{\text{tr}(P_0 |1_D\rangle\langle 1_A|)}_{=0} + \text{c.c.}]. \quad (46)$$

Using that the identity operator in the atomic subspace can be written as $|\uparrow\rangle\langle\uparrow| + |\downarrow\rangle\langle\downarrow|$, we find $F_{\text{dark}} = \frac{1}{4} \text{tr}(\rho_0^{\text{atom}})$. Using $\text{tr}(\rho_0^{\text{atom}}) = 1$, we find

$$\boxed{F_{\text{dark}} = \frac{1}{4}.} \quad (47)$$

This is plausible. The classical limit $F = \frac{1}{2}$ would be reached, if the atomic and the photonic states correlate classically with no coherence between them as, e.g., in $\rho = \frac{1}{2} |\downarrow, 1_D\rangle\langle\downarrow, 1_D| + \frac{1}{2} |\uparrow, 1_A\rangle\langle\uparrow, 1_A|$. For the dark counts, however, the atomic and photonic states are uncorrelated, which gives the worse result $F = \frac{1}{4}$.⁷

2.2 Postselected Fidelity in the Total Ensemble

The probability that an experimental shot becomes part of the postselected subensemble is

$$p(\text{ps}) = P_{\text{dark}} p_0 + P_{\text{no-dark}} p_1, \quad (48)$$

⁷Note that F can be written as $F = \frac{1}{4}(1 - S_{11} - S_{22} - S_{33})$, where S_{11} , S_{22} , and S_{33} are generalized Stokes parameters, see Eqs. (2.36) and (2.47) in Ref. [3]. With classical correlations in one basis one obtains $S_{11} = -1$ and, as the lack of coherence implies absence of correlations in the other bases, $S_{22} = S_{33} = 0$. For an uncorrelated state one finds $S_{11} = S_{22} = S_{33} = 0$.

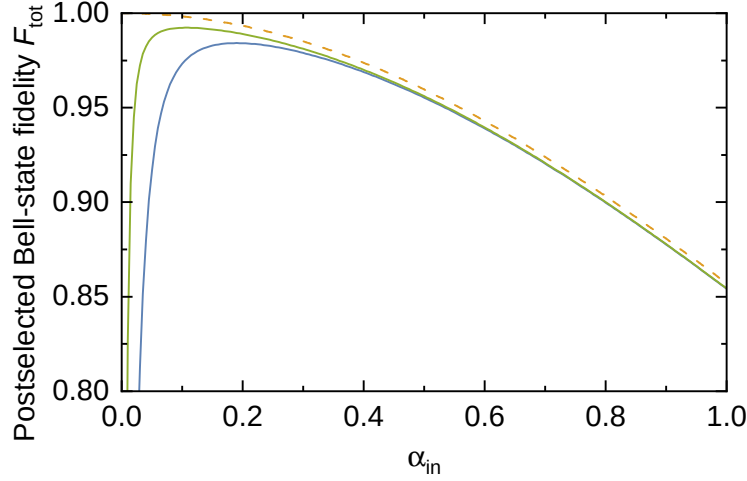


Figure 1: The postselected Bell-state fidelity F_{tot} for the parameters of table 1. The green (blue) line is for $\mu = 10^{-5}$ ($\mu = 10^{-4}$). The maximum $F_{\text{tot}} = 0.992$ ($F_{\text{tot}} = 0.984$) is reached for $\alpha_{\text{in}} = 0.11$ ($\alpha_{\text{in}} = 0.19$), corresponding to $|\alpha_{\text{in}}|^2 = 0.011$ ($|\alpha_{\text{in}}|^2 = 0.036$). The model takes into account multi-photon events for an incoming coherent state with amplitude α_{in} , combined with loss of photons in and after the cavity. It also takes into account detector dark counts. The model ignores all other experimental imperfections, e.g., interaction-time broadening, optical mode matching, imperfections in the detector polarization basis selection, imperfections in the atomic-state readout, imperfections in the initial-state preparation, unintentional detunings, laser phase noise, position dependent differential trapping potentials, etc. The dashed line represents the naive expectation $\frac{1}{2}(1 + V_{\text{det}})$ of Eq. (37) for $d_{\uparrow} = -d_{\downarrow}$, $\theta = 0$, and the absence of dark counts. V_{det} is given by the simple analytic expression (20).

where $P_{\text{no-dark}}$ is the probability that no dark count occurs. Assuming a Poisson distribution for the number of dark counts in the detection window with mean value μ , we find

$$P_{\text{no-dark}} = e^{-\mu}, \quad P_{\text{dark}} = \mu e^{-\mu}. \quad (49)$$

The conditional probability that a single detection event was a dark count is

$$p(\text{dark}|\text{ps}) = \frac{P_{\text{dark}}p_0}{P_{\text{dark}}p_0 + P_{\text{no-dark}}p_1} = \frac{\mu e^{-\mu}p_0}{\mu e^{-\mu}p_0 + e^{-\mu}p_1} = \frac{\mu p_0}{\mu p_0 + p_1}. \quad (50)$$

The total density matrix for the postselected subensemble is the weighted sum

$$\rho_{\text{tot}} = p(\text{dark}|\text{ps})\rho_{\text{dark}} + [1 - p(\text{dark}|\text{ps})]\rho_{\text{ps}}. \quad (51)$$

As the postselected fidelity is linear in ρ , we find that the postselected Bell-state fidelity including detector dark counts is the corresponding weighted sum

$$F_{\text{tot}} = \langle \psi_{\text{id}} | \rho_{\text{tot}} | \psi_{\text{id}} \rangle = p(\text{dark}|\text{ps}) \underbrace{\langle \psi_{\text{id}} | \rho_{\text{dark}} | \psi_{\text{id}} \rangle}_{=F_{\text{dark}}} + [1 - p(\text{dark}|\text{ps})] \underbrace{\langle \psi_{\text{id}} | \rho_{\text{ps}} | \psi_{\text{id}} \rangle}_{=F_{\text{ps}}}. \quad (52)$$

Hence

$$F_{\text{tot}} = \frac{\mu p_0 F_{\text{dark}} + p_1 F_{\text{ps}}}{\mu p_0 + p_1}. \quad (53)$$

This is shown in Fig. 1 for the parameters of table 1.⁸ For these parameters, we find

$$\left| \frac{r_{\uparrow}}{\alpha_{\text{in}}} \right|^2 = 0.36, \quad \left| \frac{r_{\downarrow}}{\alpha_{\text{in}}} \right|^2 = 0.49, \quad L_{\text{gen}} = 0.35, \quad L_{\text{tot}} = 0.51. \quad (54)$$

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⁸Table 2.2 of Ref. [4] quotes $\eta_{\text{esc}} = 0.85$ and $(g, \kappa)/2\pi = (142, 59.6)$ MHz for the short resonator. This value of g refers to a hypothetical scenario with the cycling transition $|5S_{1/2}, 2, 2\rangle \rightarrow |5P_{3/2}, 3, 3\rangle$ (which is circularly polarized and, hence, not an eigenmode of the resonator with non-degenerate linearly polarized eigenmodes) and a point particle fixed at the position of maximum coupling. A measurement on p. 80 of Ref. [4] gave $g/2\pi = 43.8$ MHz. This is expected to be a factor $1/\sqrt{3}$ smaller than above because it was measured for the π -polarized $|5S_{1/2}, 2, 2\rangle \rightarrow |5P_{3/2}, 3, 2\rangle$ transition. The remaining factor of ≈ 2 is most likely caused by the finite temperature of the atom in the experiment. Ref. [5] quotes $g/2\pi = 21.9$ MHz for the π -polarized $|5S_{1/2}, 2, 1\rangle \rightarrow |5P_{3/2}, 2, 1\rangle$ transition. This is compatible because the dipole matrix element is a factor of 2 smaller. Leart recently measured $g^2/2\kappa\gamma = 1.64$, corresponding to $g/2\pi = 24.3$ MHz on the π -polarized $|5S_{1/2}, 2, 0\rangle \rightarrow |5P_{3/2}, 1, 0\rangle$ transition. Its matrix element is a factor of $\sqrt{4/5} \approx 0.89$ smaller than that of Ref. [5]. The slightly better g results from cooling the atom better.